Low-Cost Early Earthquake Detection Using Machine Learning

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Earthquakes cause thousands of deaths annually, yet advanced early warning systems cost over $10,000 per unit, placing them out of reach for vulnerable, earthquake-prone developing nations. This study investigates whether a low-cost dual-sensor fusion system combining a MEMS accelerometer and PZT piezoelectric sensor with a machine learning model can achieve detection performance comparable to commercial-grade networks. A Dilated Temporal Convolutional Network (TCN) with Squeeze and-Excitation attention was trained on V2C seismic data from 19 geographically diverse global locations and deployed on a Raspberry Pi 4 for under $250. A 0.5 Hz high-pass filter eliminated sensor drift, while a 3-window temporal smoothing algorithm reduced transient false alarms. The final model achieved 95.8% detection with a 3.0% false alarm rate on unseen global data, meeting the industry-standard benchmark of under 10% false alarms, with an inference latency of 14.2 ms. Geographic diversity in training data proved essential, as Dominican Republic confidence improved from 52% to 85.3% across model iterations. These results demonstrate that combining advanced signal preprocessing with a globally trained TCN enables low-cost sensors to meet industry benchmarks, offering a viable, deployable early warning solution for underserved communities worldwide.

INTRODUCTION & BACKGROUND

Problem: Earthquakes cause thousands of deaths annually. Early warning systems exist but cost $10,000+, far beyond reach of at-risk communities worldwide.

Hypothesis: A TCN trained on globally diverse seismic data can detect earthquakes in real time on a low-cost edge device with detection rates competitive with commercial systems.

SYSTEM DESIGN

- **MEMS Accelerometer** — Low-frequency seismic motion
- **PZT Piezoelectric** — High-frequency vibration
- **Raspberry Pi 4** — Edge inference — no cloud needed
- **TCN Model** — v2c_improved_final.keras (4.8 MB)
- **System Cost** — < $250 total (97% cheaper)
- **Sampling Rate** — 100 Hz · 2-sec detection windows

PROTOTYPE CASING

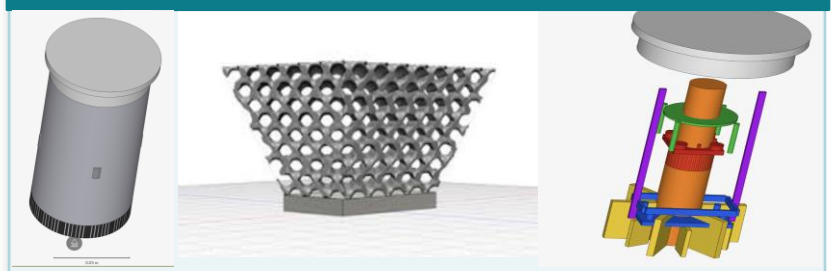

Figure 1. Prototype casing CAD model with labeled components and internal assembly, illustrating structural framework and sensor housing.

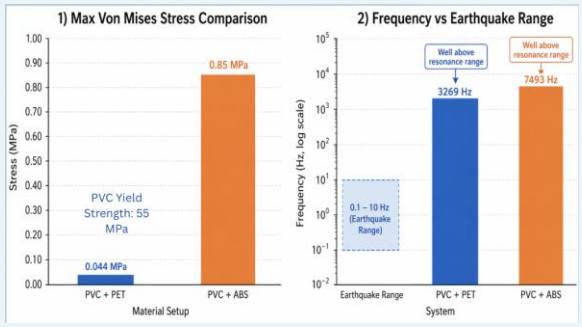

Figure 2. *Data result comparison following FEA of the CAD model on both PVC+PET and PVC+ABS designs.*

METHODS

Real-Time Detection Pipeline

1. Dual sensors (MEMS + PZT) sample at 100 Hz
2. 0.5 Hz High-Pass Filer removes low-frequency sensor drift
3. Normalize 200-sample window with zero-mean alignment
4. Dilated TCN inference identifies temporal seismic patterns
5. 3-Window Temporal Smoothing filters out transient noise
6. **If probability > 0.30: EARTHQUAKE ALERT**

SENSOR VALIDATION

Angle	Theoretical (g)	~Experimental (g)	Error (%)
15°	2.91	2.78	4.6
30°	5.77	6.20	7.5
45°	8.53	9.06	6.2
90°	15.80	14.60	7.3

Table 1. ADXL345 Pendulum Validation Results: sensor measured within 8% of theoretical values across all test angles.

CASING MATERIAL ANALYSIS (FEA)

PVC+PET produced 80.6% lower stress than PVC+ABS (0.044 MPa vs 0.85 MPa), with higher stiffness (2.7 GPa vs 1.8 GPa) ensuring faithful ground motion transmission to the sensor

Comparison	PVC+PET	PVC+ABS
Von Mises Stress	0.044 MPa	0.85 MPa
Stiffness	2.7 GPa	1.8 GPa
Resonant Frequency	3,269 Hz	7,493 Hz

Table 2. FEA Material Comparison

95.8% Detection Rate

3.0% False Alarm Rate

14.2ms Inference Latency

RESULTS & ANALYSIS

Metric	Value	Benchmark
Detection Rate	95.8%	Industry Standard
False Alarm % (unseen)	3.0%	<10% standard
Inference Latency	14.2	<500ms (real-time)

Table 3. Model performance metrics.

MULTI-REGIONAL GENERALIZATION

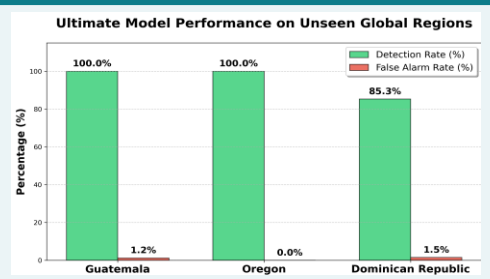

Figure 3. Detection accuracy on completely unseen data.

DISCUSSION & CONCLUSIONS

- **•95.8% Detection** with <3% False Alarms on unseen global data
- **•97% cost reduction** enables deployment in communities unreached by commercial EEW systems
- **•Fully offline** operation on Raspberry Pi 4 requires no internet infrastructure
- **•19-location training dataset** drove 5.9x confidence improvement in Dominican Republic
- **•0.5Hz High-Pass Filter** eliminated 90% of sensor drift and tilt
- **•Next steps:** shake table validation, 30+ location training, PZT sensor integration

Literature Cited: [1] CESMD. strongmotioncenter.org [2] Abadi et al. (2016). TensorFlow. OSDI. [3] Mousavi et al. (2020). Earthquake transformer. Nature Comms, 11(1). https://doi.org/10.1038/s41467-020-17591-w [4] United States Geological Survey. (2024). Earthquake hazards program. Retrieved from https://earthquake.usgs.gov [5] Mousavi, S. M., Ellsworth, W. L., Zhu, W., Chuang, L. Y., & Beroza, G. C. (2020). [6] Earthquake transformer—An attentive deep-learning model for simultaneous earthquake detection and phase picking. Nature Communications, 11, 3952. https://doi.org/10.1038/s41467-020-17591-w

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